American Computer Science League

2025-2026 ● Contest 4: Short Problem Solutions ● Elementary Division

1. Graph Theory

Given that A is Red, it's possible for B and E to be green. Then C and D must be different. However, if B and C are Green, then E and D can be Blue. There must be at least 3 since AEC is a cycle. The 4-color Theorem proves that there never need to be more than 4 colors on any map in the world.

C. 3

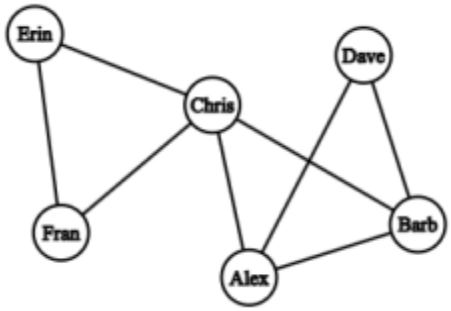
2. Graph Theory

Eliminating the edge from Dubhe to Megrez makes 2 odd vertices even. Then starting with Alkaid and ending with the unnamed star at the bottom of the Little Dipper's square, the graph can be traced without lifting your pencil.

A. Dubhe to Megrez

3. Graph Theory

The graph looks like the following:



After drawing the graph, there are 3 different cycles of 3 people: ABC, CEF, ABD.

D. 3

4. Graph Theory

By inspection, the possible routes are MPR, MPSR, MTSR, MTSPR, MTNPR, MPNTSR, MTNPSR.

E. 7

5. Graph Theory The possibilities are ABCF, AEDF, ABCDF, AEDCF, but the shortest distance is 15 miles with 3 edges.	C. 15 miles
6. Graph Theory The images are similar, but the edges are not the same in any of them. Graph #1 has the edges {AB, AD, CD, BC}. Graph #2 has edges {AC, BC, AD,BC}. Graph #3 has edges {AB, BD, AC, CD}.	E. None of them